

# PROTEIN STRUCTURE PREDICTION

## Objective

To predict and visualize the three-dimensional (3D) structure of a protein using AlphaFold DB and PyMOL.

## Protein Information

Protein: Myoglobin (P02144)

Organism: Homo sapiens

Prediction Tool: AlphaFold DB

Visualization Tool: PyMOL

## Structural Features

Myoglobin is a monomeric oxygen-binding protein.

The structure is predominantly composed of alpha-helices.

AlphaFold predicted the structure with very high confidence (average pLDDT  $\approx$  97).

The protein exhibits a compact globular fold important for oxygen storage in muscle tissue.

## Conclusion

The 3D structure of human Myoglobin was successfully predicted using AlphaFold DB and visualized using PyMOL. The predicted model showed high confidence and revealed key structural features associated with oxygen storage and transport in muscle tissues.